

MODEL QUESTION PAPER SET I**Medical Imaging Technique**

Time: 3 Hour

Max. Marks: 75

PART A**I. Answer all the following questions****(9 x 1 = 9 Marks)**

1. The CT number of water is -----
2. The diagnostic frequency range of ultrasound is -----
3. The FID stands for -----
4. The cryogenic agent used in superconducting magnet is -----
5. In which order do gamma rays fall on the sections in a gamma camera
 - (a) collimator, crystal, photomultiplier
 - (b) crystal, collimator, photomultiplier
 - (c) photomultiplier, collimator, crystal
 - (d) any of the above
6. Angioplasty is a ----- technique.
 - (a) Diagnostic
 - (b) Therapeutic
 - (c) Analytical
 - (d) None of the above.
7. The rotating anode in an X-ray tube -----
 - (a) increases tube life
 - (b) increases efficiency
 - (c) decreases efficiency
 - (d) Both (a) and (b)

The more the difference in the acoustic impedance in a tissue interface, the ----- the reflection will be.

- (a) More
 - (b) lesser
 - (c) minimum
 - (d) None of the above

One type of functional imaging modality is

- (a) X-ray
 - (b) CT
 - (c) PET
 - (d) MRI

PART B**II. Answer any *eight* questions from the following. Each question carries 3 marks.****(8x 3= 24 Marks)**

1. Describe the working of a fixed anode X-ray tube.
2. Explain any one type of X-ray detector.
3. Differentiate between angiography and angioplasty.
4. Discuss the modes of operation of ultrasound scanner.
5. Compare and contrast sound waves and electromagnetic waves.

6. For an ultrasound probe with frequency 4Mhz and speed in the medium is 1400m/s. Find out the wavelength and specify which form of interaction will be dominating in a matter with particle size less than wavelength.
7. Specify the function of RF coils in MRI machines.
8. List the properties of Gamma rays.
9. Explain the working of pulse height analyzer.
10. The half-life of Tc-99m is 6 hours. For nuclear imaging if a 10g of Tc-99m is injected what will be the amount of isotope available after 24 hours.

PART C

Answer ALL questions. Each question carries 7 marks.

III. Describe the working of an X-ray machine with the help of a block diagram.

OR

IV. Explain the block diagram of the fluoroscopy system.

V. Discuss the working principle of a CT machine.

OR

VI. Illustrate the CT machine generations.

VII. Explain the working of echocardiograph.

OR

VIII. With the help of block diagram explain the working of pulse echo machine.

IX. Summarize the working principle of MRI machine.

OR

X. Define the terms

(a) T1 relaxation time (2)

(b) T2 relaxation time (2)

(c) Larmor frequency (3)

XI. Describe the working of a MRI machine with block diagram.

OR

XII. Summarize the role of the following parts in an MRI machine.

(a) Magnet (4)

(b) Gradient coils (3)

XIII. Illustrate the working principle and instrumentation of a PET machine.

OR

XIV. What is Gamma camera? Explain its working.

MODEL QUESTION PAPER SET II**Medical Imaging Technique**

Time: 3 Hour

Max. Marks: 75

PART A

I. Answer all the following questions

(9 x 1 = 9 Marks)

1. Select the correct order of waves on the EMS
 - (a) Radio, Micro, Infra, Visible, Ultra, X-ray, Gamma
 - (b) Gamma, X-ray, Ultra, Visible, Infra, Micro, Radio
 - (c) Visible, Micro, Infra, Ultra, X-ray, Gamma, Radio
 - (d) Micro, Radio, Visible, Infra, Ultra, X-ray, Gamma

2. A ----- is a longitudinal wave that is caused by vibrations and travels through a medium.
 - (a) light wave
 - (b) sound wave
 - (c) X-ray
 - (d) electromagnetic wave

3. An advantage of MRI over CT Scan is its greater contrast resolution for soft tissues.
 - (a) True
 - (b) False

4. The most preferred imaging modality for brain anatomy imaging -----
 - (a) CT
 - (b) MRI
 - (c) PET
 - (d) Ultrasound

5. The function of collimator in a gamma camera is -----
 - (a) focusing
 - (b) filtering low energy gamma rays
 - (c) cooling
 - (d) localization of gamma rays

6. What is unique about fluoroscopies?
 - (a) live imaging
 - (b) no radiation
 - (c) uses sound waves
 - (d) uses microwaves

7 This scan combines a series of X-ray images taken from different angles around your body and uses computer processing to create cross-sectional images (slices) of the bones, blood vessels and soft tissues inside your body.

- (a) X-ray imaging
- (b) CT imaging
- (c) Fluoroscopy
- (d) MRI

8 What is the Doppler Effect?

- (a) A change in frequency due to the relative motion of a wave source and an observer
- (b) When a wave overlaps with another wave
- (c) When a wave bounces off of another object
- (d) When a wave bends due to traveling through a different medium

9 The % of the parent isotope remaining after 3 Half Lives.

- (a) 50%
- (b) 25%
- (c) 12.5%
- (d) 6.25%

PART B

II. Answer any eight questions from the following. Each question carries 3 marks.

(8x 3= 24 Marks)

1. What is angiography?
2. Summarize the working of spiral CT.
3. List the properties of X-ray.
4. Describe the types of ultrasound scanners.
5. Discuss about the velocity of sound in body.
6. Low frequency ultrasound probes are used to scan abdomen. Interpret the reason.
7. What is FID signal? Discuss its importance in MRI imaging.
8. Define Half –life period of a radioisotope. Comment on the half-life radioisotopes used for medical imaging.
9. Explain the procedure of radioisotope imaging.
10. List six applications radioisotope imaging

PART C

Answer ALL questions. Each question carries 7 marks.

III. With the help of a diagram, describe the working of a rotating anode x-ray tube.

OR

IV. Illustrate the block diagram of C-arm.

V. Explain the working a CT machine with the help of a block diagram.

OR

VI. Illustrate any five generations of CT machines.

VII. Describe the construction of a ultrasound probe.

OR

VIII. Discuss the applications of A-mode scan.

IX. With the help of a block diagram explain the working of MRI machine.

OR

X. What is resonance in MRI machine? Summarize the role of RF coils in MRI machine.

XI. List the types of magnets used in MRI machine. Explain any one in detail.

OR

XII. Define T1 and T2 relaxations. Illustrate the process of T1 and T2 relaxations.

XIII. Illustrate the working principle and instrumentation of a SPECT machine.

OR

XIV. Explain the working of pulse height analyzer.